Higher National Diploma in Engineering (Mech/BSE/Elect)

First Year, Second Semester Examination - 2018

EE1208-ELECTRICAL PRINCIPLES B

Answer any 5 questions Time : 3 hours

Q1.

(a) Define the following terms related to a network graph:

(i) Directed Graph (ii) Tree (iii) Co-tree (iv) Twig

(2 marks)

(b) For the circuit shown in Fig. Q1(a):

(i) Draw the directed graph of the network (2 marks)

(ii) Draw a tree by selecting branch no: 2 and no: 4 from the oriented graph (1 mark)

(iii) Write the cut set matrix (2 marks)

(iv) Write the branch admittance matrix, $[Y_b]$ (2 marks)

(v) Write the branch current source matrix (1 mark)

(vi) Write the branch voltage source matrix (1 mark)

(vii) Write the twig voltage matrix (1 mark)

(viii) Set the node pair equations:

$[Q][Y_b][Q]^T[V_t] = [Q][i_s] - [Q][Y_b][v_s]$ Find the value for v (6 marks)

(c) For the circuit given in Fig. Q1(b), obtain the node voltages V_A, V_B and V_C by using nodal equations (Resistor values are in ohms) (6 marks).

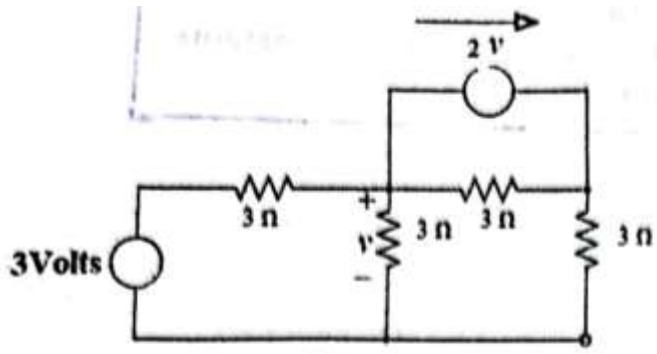

Figure. Q1(a)

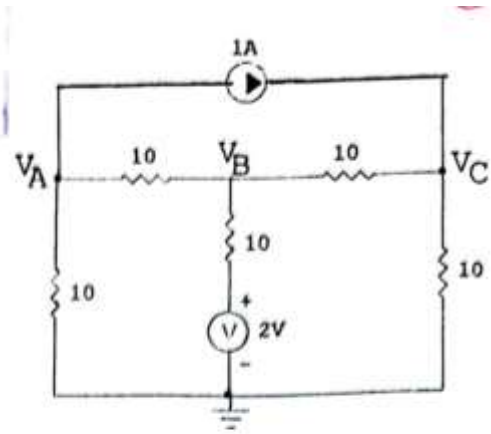

Figure. Q1(b)

Q2.

(i) State the construction difference between **single transient circuit** and **double transient circuit** (2 marks)

(ii) List down the type of **surges** occurred in the overhead transmission lines and briefly discuss the causes for generation of those surges (4 marks)

(iii) Draw the **lightning surge wave form** relevant to the data given below: $10\mathrm{kA}$, $2.5\mu\mathrm{sec}$, $25\mu\mathrm{sec}$ (3 marks)

(iv) A $50\mathrm{Hz}$ sinusoidal voltage of maximum value $100\mathrm{volts}$ is applied to a series circuit of resistance 5Ω and inductance of $0.2\mathrm{H}$. Find an expression for the value of the current at any instant after the voltage is applied, assuming that voltage is zero at the instant of application. Calculate the current value 0.05 second after switch on (6 marks)

(v) The R-L-C series circuit contains a coil of inductance $L = 423\mathrm{mH}$, resistance $R = 80\ \Omega$ and capacitance $C = 169.2\ \mu\mathrm{F}$. A step voltage of $2\mathrm{V}$ is applied to the circuit. Assume that current $i = 0$ at the instant of applying the step voltage. Then determine: a. the state of damping (1 mark) b. an expression for current i (3 marks) c. Is it possible to obtain critical damping for above circuit? If so, how do you obtain critical damping? (1 mark)

Q3.

(a) (i) The critical frequency of a **low pass active filter** is f_c. Mark the **Bandwidth** of the filter on filter response curve (2 marks). (ii) What is the benefit of increasing the **roll off rate** of active low pass filter? (1 mark)

(b) (i) Design a **high pass passive filter** having a cut-off frequency of 2kHz with a load resistance of $300\ \Omega$ in: (i) T - configuration (ii) π - configuration (4 marks)

(ii) An active low pass filter network having a roll off rate of -20dB/decade/pole. Draw the suitable active filter network in which the roll off rate increase to -60dB/decade/pole (3 marks)

(iii) Determine the **nominal impedance** and the **cut off frequency** of filter circuits given in figure Q3(a) and figure Q3(b) (10 marks)

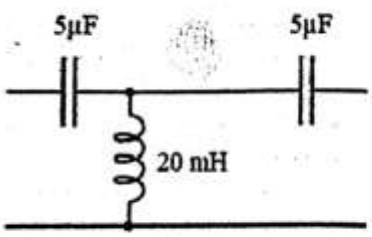

Figure.Q3(a)

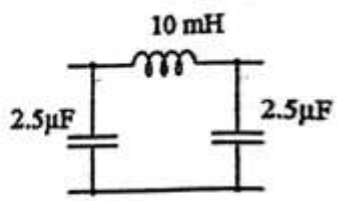

Figure.Q3(b)

Q4.

(a)

(i) Write the voltage equation at any distance of the mismatched transmission line (1 mark).

(ii) What is the correctly terminated transmission line? (1 mark)

(iii) Define the following terms related to the mismatched transmission line:

(I) Reflection coefficient

(II) Standing wave ratio, SWR (3 marks)

(iv) A transmission line has a characteristic impedance of $10\ \Omega$ and negligible loss. If the terminating impedance of the line is $(6 + j4)\ \Omega$, determine: (I) the **reflection coefficient** (II) the **standing wave ratio (SWR)** (3 marks)

(b) (i) Draw the two equivalent circuits when $\lambda/4$ line fulfill the following conditions: (I) Length less than $\lambda/4$, terminating end is short circuit and other end is open circuit (II) Length less than $\lambda/4$, terminating end is open circuit and other end is also open circuit (2 marks) (ii) State the **demerit of single stub machine** in transmission line (2 marks) (iii) Briefly explain the method to be adopted to overcome the disadvantage of single stub machine (2 marks) (iv) A cable has the following primary constants:

- Resistance, $R = 80\ \Omega/\text{loop km}$
- Conductance, $G = 2\ \mu\text{S}/\text{loop km}$
- Capacitance, $C = 5\ \text{nF}/\text{loop km}$ Determine the following for minimum distortion at a frequency of 1.0kHz: (I) the value of inductance per loop km required (II) Characteristic Impedance, Z_0 (III) Propagation coefficient, γ (IV) attenuation coefficient, α and Phase delay, β (V) Wavelength, λ on the line (VI) velocity of propagation of signal (6 marks)

Q5.

(i) List out the two types of **optical sources** commonly used in the fiber optic communication and briefly describe the materials used for their construction (4 marks)

(ii) Briefly explain the process of **data transmission through fiber optics** (3 marks)

(iii) What is the condition to be fulfilled by the **refractive indices of core and cladding** of the fiber optic cable for successful data transmission (2 marks)

(iv) Briefly explain why **optical Bandwidth is larger than that of the electrical Bandwidth** of same capacity (3 marks)

(v) State **3 losses in fiber optics** (3 marks)

(vi) Standard size of step index fiber in cable based is given as:

- Diameter of core $= 50\ \mu\text{m}$
- Diameter of cladding $= 125\ \mu\text{m}$ The typical value of core and cladding refractive indexes are 1.5 and 1.49 respectively. Determine: (I) **Numerical Aperture (NA)** (II) **Fractional difference in the refractive index** (III) The **velocity of light in the optically active region of the material** (5 marks)

Q6. Electromagnetic Waves and Energy Conversion

(a) (i) State the three **mechanisms used to transmit radio waves** from transmitting antenna to receiving antenna (3 marks) (ii) State the types of information that can be obtained through the **pointing vector** (2 marks)

(b) The electric field of a travelling electromagnetic wave is given by $E(z,t) = 10\ \hat{a}_x \cos(\pi \times 10^7 t + \pi z/15)$ and the intrinsic impedance of the medium is $98.7\ \Omega$. Determine: (i) The **direction of wave propagation** (ii) The **wave frequency**, f (iii) The **phase change coefficient**, β (iv) Its **wavelength**, λ (v) Its **phase velocity**, v_P (vi) **Ratio between relative permittivity and permeability** of the medium (vii) The **time average power carried by the wave** (8 marks)

(c) (i) Briefly explain the operation of a simple **electromagnetic relay** in relation to **energy transfer** in this system (3 marks) (ii) Write the **energy balance equation for the motor action** (2 marks)

(d) (i) State the main **advantage of electromechanical energy** (2 marks) (ii) Why do you think that the **heat energy generated by either motor action or generator action is irreversible conversion**? (2 marks)